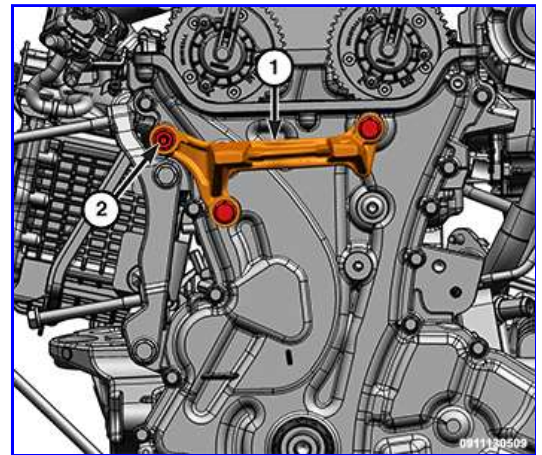


EC1,EC3/ Removal And Installa...**REMOVAL AND INSTALLATION**

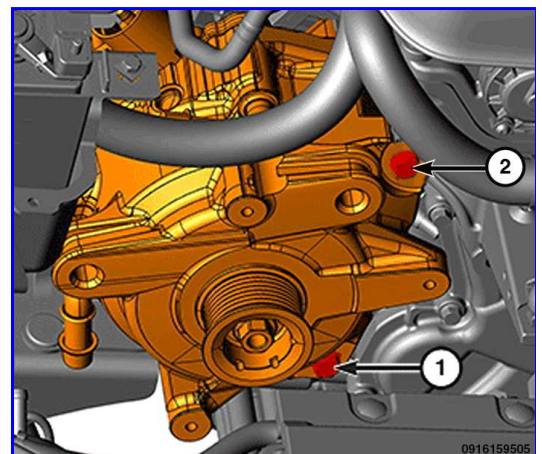
Special Tools: Click to display a list of tools used in this procedure

REMOVAL

1. Perform the fuel system pressure release procedure
([Refer to 14 - Fuel System/Fuel Delivery/Standard Procedure](#)).
2. Raise and support the vehicle ([Refer to 04 - Vehicle Quick Reference/Hoisting/Standard Procedure](#)).
3. Disconnect and isolate the negative battery cable. If equipped with an Intelligent Battery Sensor (IBS), disconnect the IBS connector first before disconnecting the negative battery cable.
4. Remove the cylinder head cover
([Refer to 09 - Engine/Cylinder Head/COVER\(S\) , Cylinder Head/Removal and Installation](#)).
5. Remove the electric vacuum pump
([Refer to 05 - Brakes, Base/Electrical/PUMP, Electric Vacuum/Removal and Installation](#)).
6. Remove the bolts (2) and the electric vacuum pump mounting bracket (1).



7. If equipped with eTorque, remove the eTorque belt tensioner
([Refer to 07 - Cooling/Accessory Drive/TENSIONER, Belt/Removal and Installation](#)).
8. If equipped with eTorque, remove the upper bolt (2), loosen the lower bolt (1) and reposition the Motor Generator Unit (MGU) away from the timing cover.

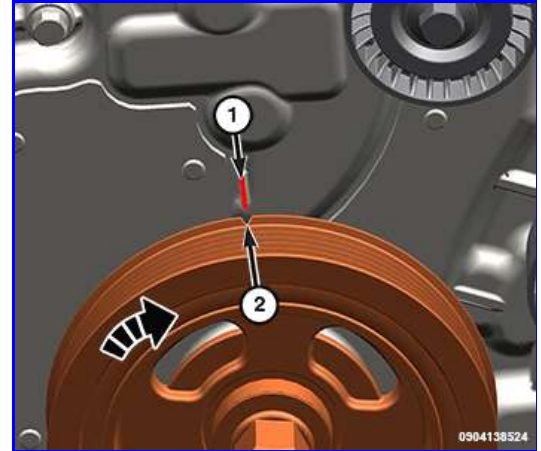


9. Remove the accessory drive belt
([Refer to 07 - Cooling/Accessory Drive/BELT, Serpentine/Removal and Installation](#)).
10. Remove the accessory drive belt idler pulley
([Refer to 07 - Cooling/Accessory Drive/PULLEY, Idler/Removal and Installation](#)).

11. Remove the accessory drive belt tensioner

(Refer to 07 - Cooling/Accessory Drive/TENSIONER, Belt/Removal and Installation).

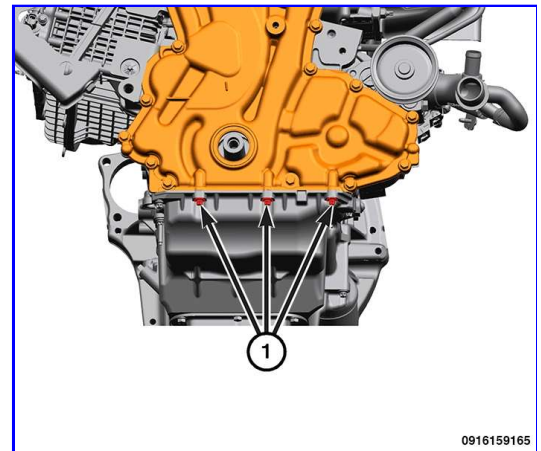
12. Rotate the crankshaft clockwise (as viewed from the front). Align the timing cover mark (1) with the damper notch (2) to place the number one cylinder piston at top-dead-center (TDC) on the exhaust stroke.



13. Remove the vibration damper

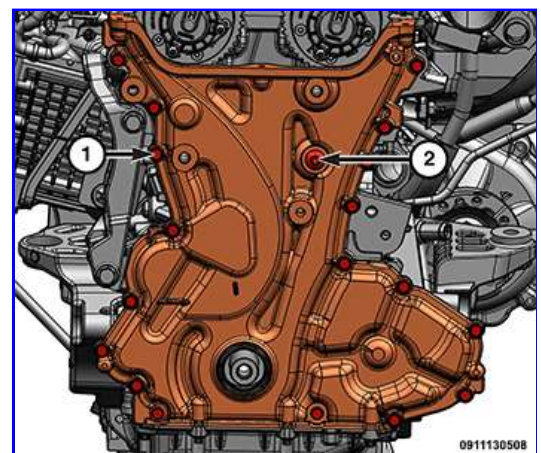
(Refer to 09 - Engine/Engine Block/DAMPER, Vibration/Removal and Installation).

14. Remove the three oil pan to timing chain cover bolts (1).



15. Remove the bolt (2) from the timing cover.

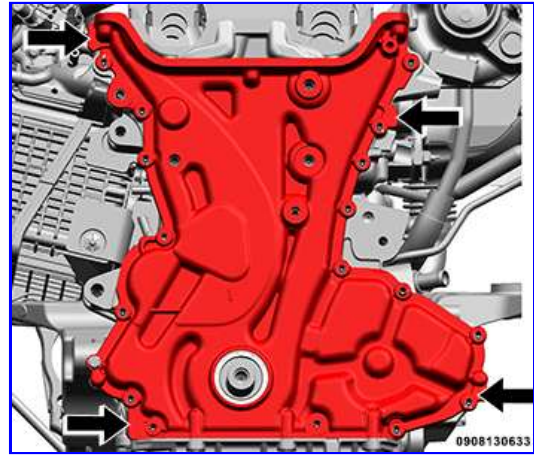
16. Remove the eighteen timing cover bolts (1).



NOTE: It may be necessary to warm the timing chain cover/oil pan interface with a heat gun to help loosen the sealant joint.

17. Use a suitable tool at the indicated pry points to release the timing chain cover from the sealant. To avoid bending the timing chain cover, release the lower flange first then release the upper flange.

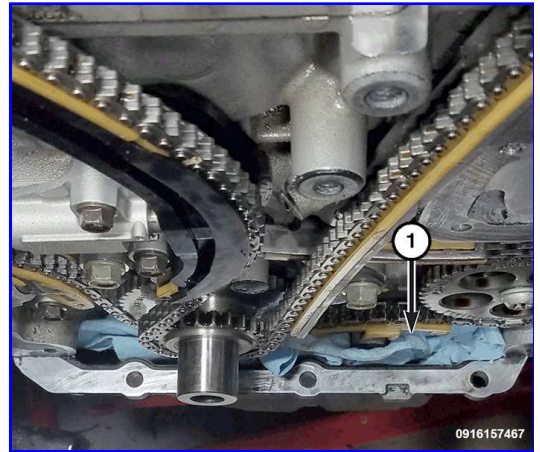
18. Remove the timing cover from the locating dowels and the engine crankshaft.



CAUTION:

Do not use oil based liquids, wire brushes, abrasive wheels or metal scrapers to clean the engine gasket surfaces. Use only isopropyl (rubbing) alcohol, along with plastic or wooden scrapers. Improper gasket surface preparation may result in engine fluid leakage.

19. Place a protective cover (1) in the oil pan to prevent debris from entering the engine.



20. Remove the sealant residues from the oil pan, engine block, cylinder head and timing chain cover [\(Refer to 09 - Engine/Standard Procedure\)](#).

21. If required, remove the front crankshaft oil seal from the timing chain cover.

NOTE:

Be careful not to damage the timing chain cover when removing the seal.

22. Check the integrity of the timing chain cover locating dowels.

INSTALLATION

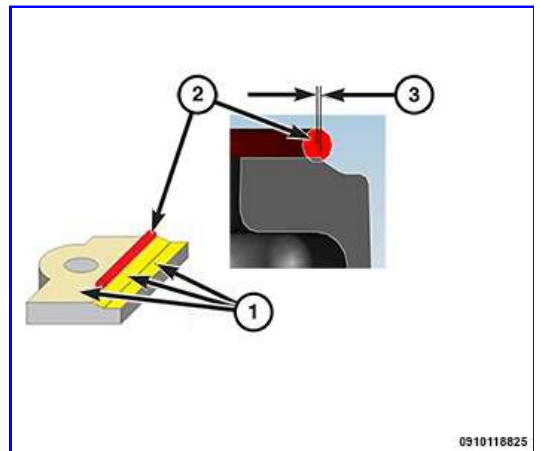
1. Clean the engine timing cover (1), cylinder head and cylinder engine block mating surfaces with isopropyl alcohol in preparation for sealant application.

CAUTION:

Engine assembly requires the use of a unique sealant that is compatible with engine oil. Using a sealant other than Mopar® Threebond Engine RTV Sealant may result in engine fluid leakage.

CAUTION:

Following the application of Mopar® Threebond Engine RTV Sealant to the gasket surfaces, the components must be assembled within 10 minutes and the attaching fasteners must be

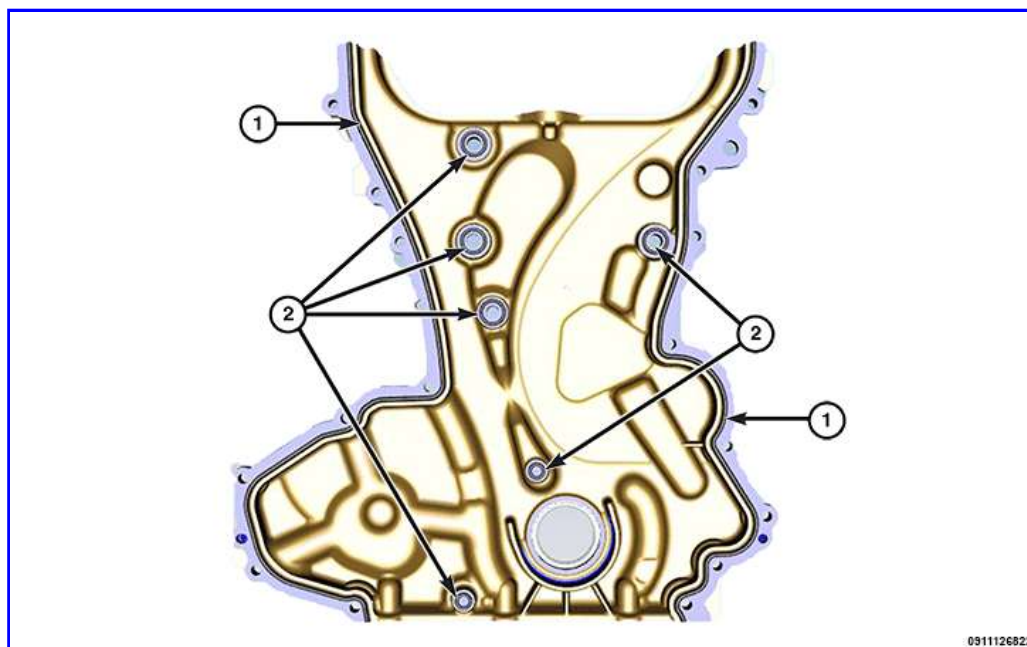


tightened to specification within 45 minutes. Prolonged exposure to the air prior to assembly may result in engine fluid leakage.

2. Apply a 3 to 4 mm bead of Mopar® Threebond Engine RTV Sealant (2) to the timing chain cover flange (1). Apply the bead of sealant 0 - 2 mm from the edge of the chamfer (3).

NOTE:

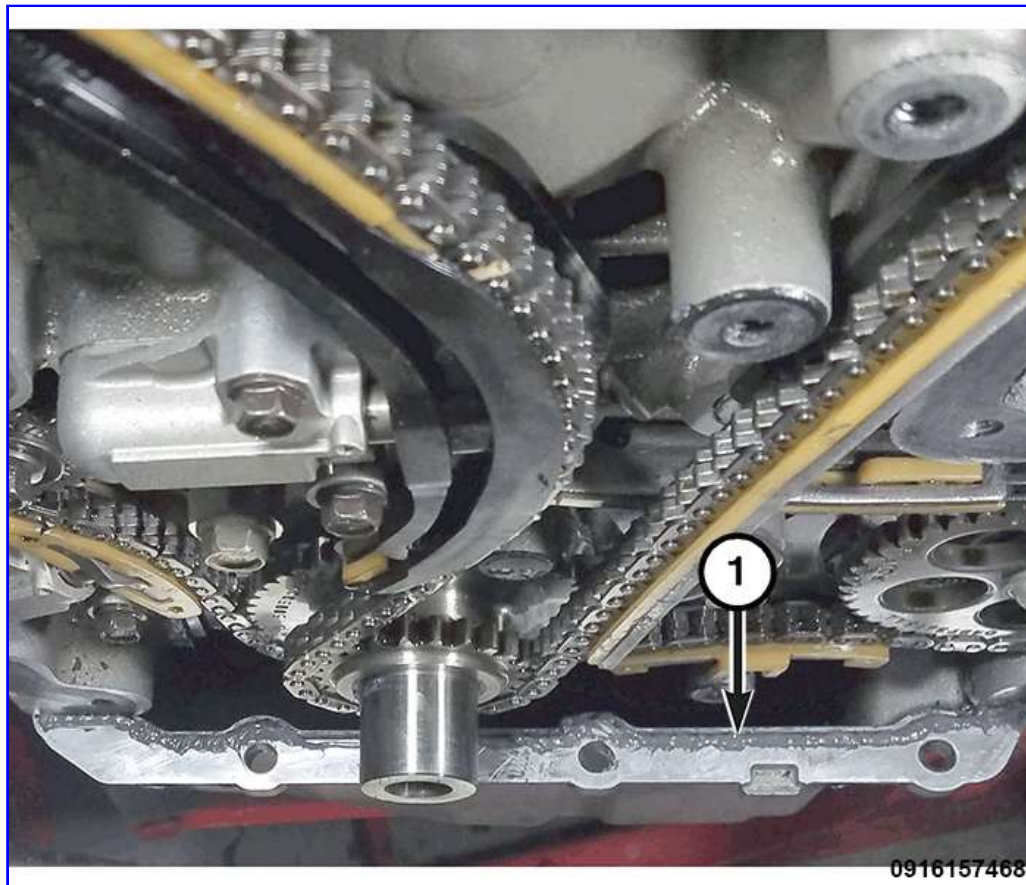
The bead of sealant must be uninterrupted.



3. Apply a 2 to 3 mm bead of Mopar® Threebond Engine RTV Sealant to the perimeter of the six locations (2) indicated in the central zone of the timing chain cover.

NOTE:

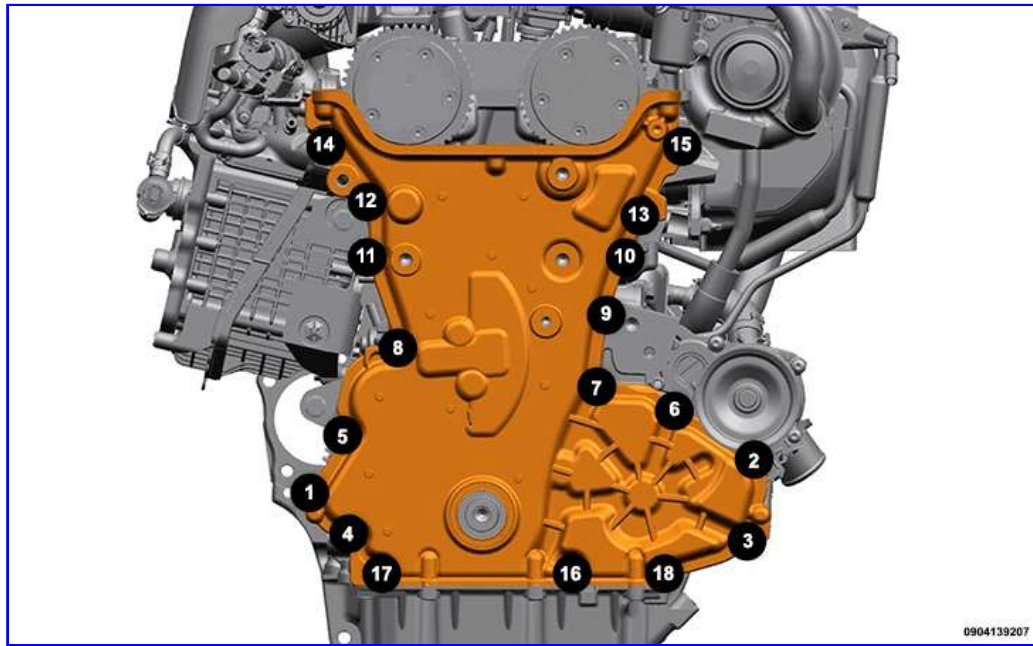
The bead of sealant must be uninterrupted.



NOTE: Remove the protective cover from the oil pan prior to timing chain cover installation.

4. Apply a 2 to 3 mm bead of Mopar® Threebond Engine RTV Sealant to the oil pan flange as shown.

NOTE: The bead of sealant must be uninterrupted.

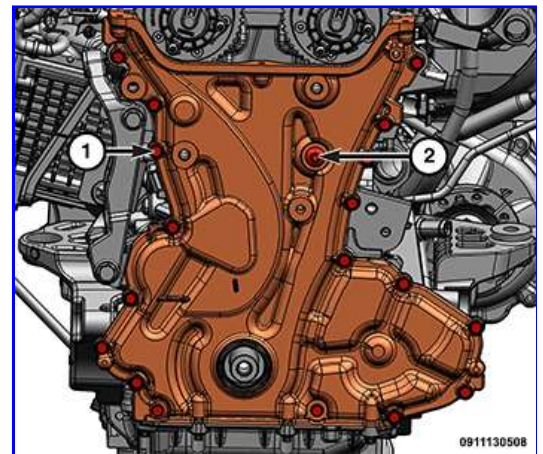


5. Keeping the timing chain cover in a vertical position with a slight outward tip at the top of the cover, load the timing chain cover at the bottom, along the oil pan and rock the top of the cover into place, being extremely careful not to wipe or disturb the Mopar® Threebond Engine RTV Sealant pattern.

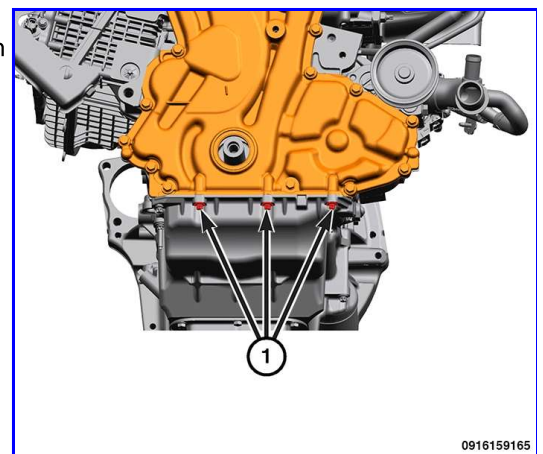
6. Align the locator pins on the engine block to the timing chain cover.

7. Install the timing chain cover and tighten the eighteen bolts in the sequence shown to the proper [\(Refer to 09 - Torque Specifications\)](#).

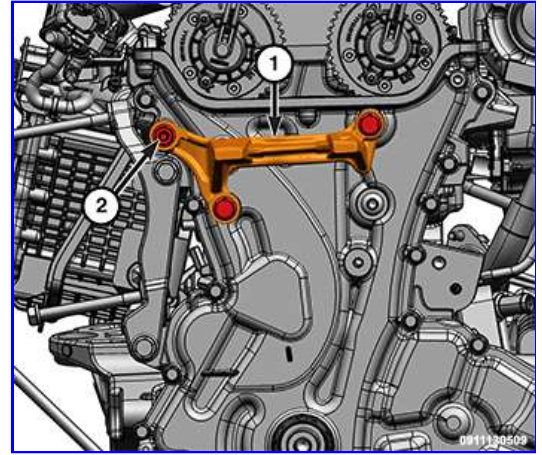
8. Install the bolt (2) to the timing cover and tighten to the proper [\(Refer to 09 - Torque Specifications\)](#).



9. Install the three oil pan to timing chain cover bolts (1) and tighten to the proper [\(Refer to 09 - Torque Specifications\)](#).

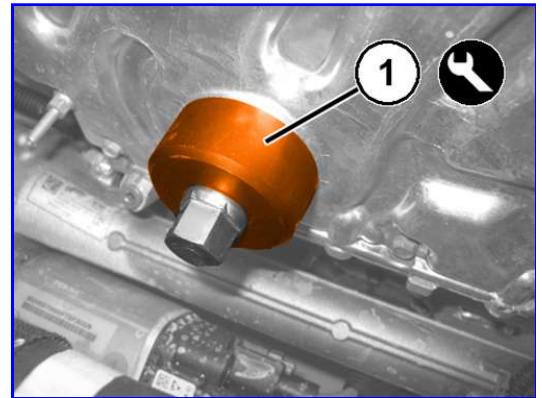


10. Install the electric vacuum pump mounting bracket (1) with three bolts (2) tightened to the proper
(Refer to 09 - Torque Specifications).



11. Install the accessory drive belt idler pulley
(Refer to 07 - Cooling/Accessory Drive/PULLEY, Idler/Removal and Installation).

12. If removed, install a **NEW** front crankshaft oil seal in the timing chain cover using [Installer, Oil Seal 9506](#) (1).



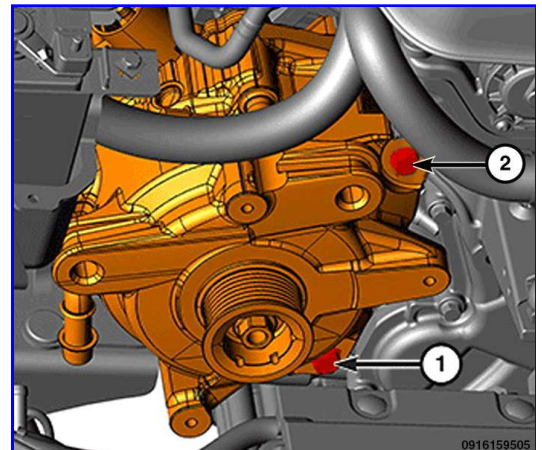
13. Install the vibration damper
(Refer to 09 - Engine/Engine Block/DAMPER, Vibration/Removal and Installation).

14. Install the accessory drive belt tensioner
(Refer to 07 - Cooling/Accessory Drive/TENSIONER, Belt/Removal and Installation).

15. Install the accessory drive belt
(Refer to 07 - Cooling/Accessory Drive/BELT, Serpentine/Removal and Installation).

16. If equipped with eTorque, install the MGU upper bolt (2). Tighten the MGU upper bolt (2) and lower bolt (1) bolt to the proper
(Refer to 08 - Torque Specifications).

17. If equipped with eTorque, install the eTorque belt tensioner and drive belt



(Refer to 07 - Cooling/Accessory Drive/TENSIONER, Belt/Removal and Installation).

18. Install the cylinder head cover
(Refer to 09 - Engine/Cylinder Head/COVER(S), Cylinder Head/Removal and Installation).

19. Install the electric vacuum pump

[\(Refer to 05 - Brakes, Base/Electrical/PUMP, Electric Vacuum/Removal and Installation\)](#).

20. If required, change the engine oil and filter

[\(Refer to 09 - Engine/Lubrication/FILTER, Engine Oil/Removal and Installation\)](#).

21. Connect the negative battery cable.

22. Remove the support and lower the vehicle.

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